RAJALAKSHMI ENGINEERING COLLEGE

B Priyanka 251001200-2025-IT

B2

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|---|---|
| **Started on** | Tuesday, 18 August 2026, 9:17 AM |
| **State** | Finished |
| **Completed on** | Tuesday, 18 August 2026, 9:28 AM |
| **Time taken** | 10 mins 47 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

**Note:** No need of counter increment for declarations and scanf() and counter variable printf() statement.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**Answer:**

```c
#include <stdio.h>

void Factor(int num)
{
    int count = 0;

    for(int i = 1; i <= num; i++)
    {
        count++;                    // i <= n

        count++;                // if condition

        if(num % i==0)                  // printf("%d ", i)
        {
            count++;                // if condition
        }         // i++
    }

    count++;                        // final i <= n

    printf("%d", count);
}

int main()
{
    int n;
    scanf("%d", &n);

    Factor(n);

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 12 | 31 | 31 | ✔ |

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 25 | 54 | 54 | ✔ |
| ✔ | 4 | 12 | 12 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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